

Rashmi Rama Sushil

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SUMMARY

Quantitative finance professional with 3 years building and operating risk analytics frameworks at Morgan Stanley Trading Risk Management, now leading the development of an AI-assisted commodity risk dashboard using GitHub Copilot and agent-based AI tooling. Daily exposure to a large, diverse commodity derivatives book spanning crude oil, EU power and natural gas, precious and base metals (LME/CME), agricultural markets, and Asian-listed futures (DCE/SGX). MS Research from IIT Madras (CGPA 9.56/10) with 4 peer-reviewed publications. CFA Level 1 certified. Combines derivatives domain expertise with hands-on AI-assisted software engineering to modernise how risk is analysed, visualised, and communicated to stakeholders.

SKILLS

Technical: Python (pandas, NumPy, SciPy, statsmodels, matplotlib), SQL, Power BI, Excel, Bloomberg Terminal, GitHub Copilot & Agent-Based AI Development (VS Code), Claude, Cursor
Finance: Derivatives Analytics, Greeks (Delta/Gamma/Vega/Theta), P&L Attribution, Risk Sensitivities, Factor Analytics (Beta), VaR, Stress Testing, Deal Structure Evaluation, Financial Modelling
Domain: Commodity Derivatives – crude oil, EU power & natural gas, precious & base metals (LME/CME), agricultural, Asian futures (DCE/SGX); Multi-asset Futures & Options

EXPERIENCE

Senior Associate, Trading Risk Management
Morgan Stanley

Jan 2025 – Present
Mumbai, India

- Leading the design and development of an AI-assisted commodity risk dashboard – using GitHub Copilot and agent-based AI tooling in VS Code to modernise how the desk visualises forward curves, time spreads, cross-curve basis spreads, and baskets of trades.
- Designing a framework for evaluating potential deal structures, systematically comparing structuring alternatives against their risk and P&L profile.
- Built and operated P&L production and reporting tools; led daily/weekly risk and P&L review forums with Director/MD-level leadership.
- Performed return attribution and derivatives exposure analytics (Greeks, stress analysis, Sharpe ratio) across multi-asset books with over \$1B in peak notional monitored.
- Developed executive dashboards and risk narratives for Director/MD-level stakeholders, translating quantitative analysis into clear, decision-relevant communication.
- Monitored 20+ instruments daily using Bloomberg to analyse price shocks, vol shifts, and macro catalysts; partnered with traders, quants, and data teams to translate stakeholder requirements into analytical tools.

Associate, Trading Risk Management
Morgan Stanley

Jul 2023 – Dec 2024
Mumbai, India

- Built, maintained, and extended 30+ structured risk reporting tools monitoring \$50M–\$300M notional across 5+ global desks; full platform ownership.
- Developed and validated quantitative models tracking derivatives risk sensitivities (Delta, Vega, Gamma, Theta).
- Built factor analytics (Beta, Hedge ratio) to identify directional risk and position concentration.
- Built structured, maintainable automation tools using Python, SQL, Power BI, and Excel – reduced

manual workload by 40%.

- Streamlined reporting and reconciliation infrastructure by mapping data lineage and integrating multiple data sources into a scalable, reusable pipeline.

PROJECTS

Beta Benchmarking

Python, live web tool

Aug 2026

GitHub

- Built an open-source study of beta's different uses across finance – CAPM, portfolio, and rolling beta for hedging, systematic risk, and cross-asset benchmarking – with a live in-browser tool to compute rolling beta on demand.

Metals Hedging Stress Test

Python, time spreads, DV01

Jun 2026

GitHub

- Stress-tested a metals derivatives hedge across front-end slippage and back-end holding risk; built liquidity screening via open interest, time-spread decomposition, rolling stress statistics, and DV01/lease-rate sensitivity for precious metals financing.

Options Risk Analytics Engine

Python, Black-Scholes, Greeks, VaR

May 2026

GitHub

- Built a Black-Scholes risk monitor computing full Greeks, historical and parametric VaR (95%/99%), and configurable stress scenarios for a 12-position options book; surfaced concentration risk missed by notional-based position sizing alone.

Options Hedging Strategies Analyser

Python, payoff analysis

May 2026

GitHub

- Built a payoff visualiser for six hedging strategies (protective put, covered call, collar, spreads, straddle) computing exact INR payoffs, breakeven points, and risk/cost trade-offs, with a live interactive tool.

EDUCATION

MS Research, Engineering Design

Indian Institute of Technology Madras

Sep 2020 – Jun 2023

Chennai, India

- CGPA: 9.56/10. Conducted research on visualising and exploring large datasets using ML models; 4 peer-reviewed publications (Springer 2025, WCSMO USA 2023, ASME USA 2022, ACSMO Japan 2022).
- Teaching assistant, independently managing and guiding 80+ students.
- Winner, ASME IDETC International Essay Competition 2022; GATE 2020 – 98.31 percentile.

ADDITIONAL EXPERIENCE

Research Intern

Indian Space Research Organisation (ISRO)

Jun 2017 – Jul 2017

India

- Analysed 20+ technical papers on inflatable antenna feasibility; built and tested prototype.

CERTIFICATIONS

CFA Level 1

CFA Institute